8 **(a)** **(i)** While the speed is constant, the direction of the velocity is changing continuously.
Since velocity is a vector, a change in direction means a change in the quantity. Therefore, the train is accelerating.

(ii) Horizontal distance between rails = $\sqrt{w^2 - E^2}$

$$\tan\theta = \frac{\text{vertical distance}}{\text{horizontal distance}} = \frac{E}{\sqrt{\left(w^2 - E^2\right)}}$$

(iii) Vertically,

$$N\cos\theta = mg \qquad -(1)$$

Horizontally, the horizontal component of normal contact force provides the centripetal force,

$$N\sin\theta = \frac{mv^2}{r} \qquad -(2)$$

$$(1)/(2)$$

$$\tan\theta = \frac{v^2}{rg}$$

$$\frac{E}{\sqrt{w^2 - E^2}} = \frac{v^2}{rg}$$

$$v = \left(\frac{Erg}{\sqrt{w^2 - E^2}}\right)^{\frac{1}{2}}$$

(b) **(i)** Passenger trains are lighter/carry a smaller weight.

(ii) CD_{max} is 110 mm for 1435 rail gauge.

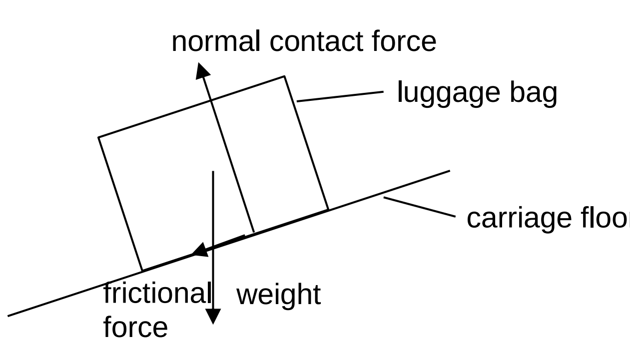
E_v = 110 + 95 = 205 mm

$$v_0 = \left(\frac{Erg}{\sqrt{w^2 - E^2}}\right)^{0.5}$$

$$= \left(\frac{\left(205\times10^{-3}\right)(1500)(9.81)}{\sqrt{1435^2 - 205^2}\times10^{-3}}\right)^{0.5}$$

$$= 46.086 = 46.1 \text{ m s}^{-1}$$

(ii) **1.**

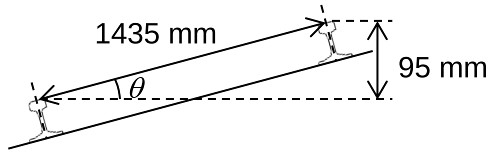


2.

$$\frac{mv^2}{r} = \frac{(19.6)\left(\frac{150000}{60 \times 60}\right)^2}{1500}$$
$$= 23 \text{ N}$$

3.

$$\sin\theta = \frac{95}{1435}$$
$$\theta = 3.8°$$



4. Vertically, $N\cos\theta = mg + f\sin\theta \quad -(1)$

Horizontally, $N\sin\theta + f\cos\theta = 23$

$\Rightarrow \quad N\sin\theta = 23 - f\cos\theta \quad -(2)$

$(2)/(1)$,

$$\tan\theta = \frac{23 - f\cos\theta}{(19.6)(9.81) + f\sin\theta}$$

$$(19.6)(9.81)\tan\theta + f\sin\theta\tan\theta = 23 - f\cos\theta$$

$f = 10.2$ N $\quad$ ($f = 9.98$ N if $F_c = 22.69$ N is used in the calculation)

Since $f < f_{max}$, luggage will not slide

OR

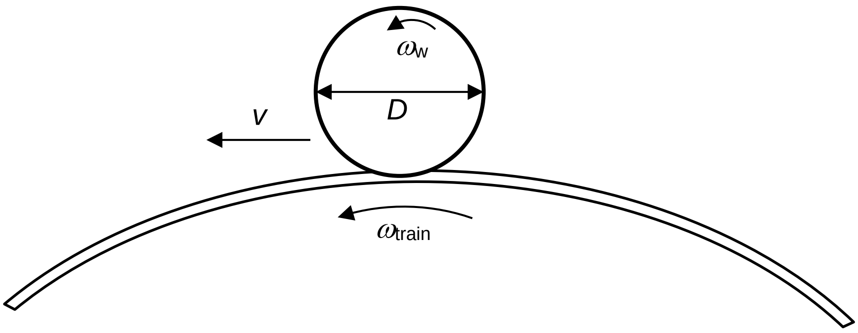
<u>Faster method</u>

the component of weight parallel to the slope and friction should provide the component of centripetal force that is parallel to the slope.

$$mg\sin\theta + f = F_c\cos\theta$$
$$(19.6)(9.81)\sin 3.8° + f = 23\cos 3.8°$$
$$f = 10.2 \text{ N}$$

(c) (i) As the circumference of the outer wheel is larger, it will be able to travel <u>a greater distance even though it rotates as the same rate</u> as the inner wheel, allowing both wheels to stay on the track.

(ii)



For a wheel of diameter D and angular velocity of ω_W moving at speed v along a curved rail of radius R, linear speed at rim of wheel = speed of wheel

$$\left(\frac{D}{2}\right)\times\omega_{\text{w}} = R\times\omega_{train}$$

Since both the inner and outer wheels are rotating with the same angular velocity of ω_{W}, and both wheels are moving along the curved rails with the same angular velocity ω_{trail},

$$\frac{D_o}{D_i} = \frac{R_{\text{outer}}}{R_{\text{inner}}}$$

$$\frac{D_o}{1.150} = \frac{200+1.524}{200}$$

$$D_o = 1.1588 = 1.16 \text{ m}$$

(d) **(i)**
- Easier maintenance (of rails)
- Cheaper or easier or faster to build (as there is no need to build tunnels or viaducts)

(ii)
- Allows land surface to be used for other purposes
- Less noise to residents / less noise pollution
- Train operations is not affected by weather conditions (rain or snow)